

Standard Operating Procedure - Cooling Water Pump Inspection

WORKFLOW AI DEMONSTRATION DATA • DUMMY DOCUMENT

1. Purpose

This SOP defines the minimum inspection and corrective-action procedure for centrifugal cooling-water pumps.

2. Safety Requirements

Apply Permit to Work where required.

Isolate electrical and mechanical energy before opening guards or coupling covers.

Use appropriate PPE including safety shoes, gloves, eye protection, and hearing protection.

Verify zero-energy state before beginning mechanical work.

3. Vibration Assessment

Normal operating vibration: below 3.5 mm/s RMS.

Alert condition: 3.5 to 4.5 mm/s RMS. Increase monitoring frequency and investigate the trend.

High condition: above 4.5 mm/s RMS. Schedule inspection at the earliest safe opportunity.

Critical condition: above 7.1 mm/s RMS with rapid increase or abnormal noise. Escalate immediately to the maintenance manager and evaluate controlled shutdown.

4. Corrective Procedure

Check foundation bolts and baseplate condition.

Inspect coupling for wear or damage.

Verify shaft and coupling alignment using approved alignment equipment.

Inspect bearing condition and lubrication.

Record vibration measurements before and after corrective work.

Return equipment to service only after safety checks and supervisor approval.

5. Documentation

All findings, measurements, corrective actions, and approval details shall be recorded in the maintenance management system.